SYNAPTIK

(Advanced Sales & Marketing Strategy)

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1- Introduction

The main purpose of this project is to “discover knowledge” by implementing a computational process to discover patterns in big data involving methods at the intersection of database systems, analytics, machine learning and artificial intelligence. Little knowledge can be gleaned from raw data, but the value of this can significantly improve with the development of data discovery techniques. This raises the ethical dilemma of how data should be treated and safeguarded.

Synaptik is to implement a variety of knowledge discovery techniques depending on the level of utilization. The goal is to search for hidden knowledge in the Synaptik selected database that we are technically capable of generating and storing. A variety of methods, depending on the Synaptik level, are available to assist in extracting patterns that when interpreted provide valuable, previously unknown, insight into the stored data. This information can be predictive or descriptive in nature. Data mining, the pattern extraction phase of our knowledge discovery process, can take on many forms, the choice dependents on the desired results. This multi-step process strategy facilitates the conversion of data to useful information.

a. Advantages

Synaptik can help technology businesses in a number of ways. The primary use is growth, indicating which parts of a business’s offerings are the most successful and which are the least. In fact, demonstrating the most potential in terms of sales and those that are a drain on the company’s resources and that may inhibit expansion.

Marketing: One of the major advantages is how conclusions reached by Synaptik can be used to develop marketing campaigns to increase sales and help grow the business. The system can be used to understand the needs, wants and questions of a specific demographic that the technology business is trying to attract. Thus, they can tailor their marketing to appeal directly to this group by providing solutions to their specific problems. In fact, the system can help marketing campaign to be more cost-effective by not attempting to appeal to people that do not intend to buy, and can influence future produce development.

Personalization: Synaptik can also lead to greater personalization opportunities for technological items by personalizing communications between company and consumers to help deepen their relationship, and ultimately engenders customer loyalty and retention.

Overall, Synaptik can be of huge benefit to technology companies in helping tailor marketing campaigns to specific groups, develop new products and enhance relationships with consumers, providing that the data gathered is used in a legal, ethical manner.

b. System Requirements

Application Framework:	*Ruby on Rails*		Data Structure Language:	*Python*
Database Computation:	*YARN*		Database Framework	*MapReduce*
Database Structure/Language:	*NoSQL*		Database Name:	*Hadoop*
Database Storage Type:	*HDFS Federation*		Database Capacity:	Unlimited

c. Data/Learning

- **Big Data**

Data have swept into every industry and business function and are now an important factor of production, alongside labor and capital. The amount of data in our world has been exploding, and analyzing it has become a key basis of competition, underpinning new waves of productivity growth, innovation, and consumer surplus in today's era. In a decision making system, the bigger is the data the higher is the likelihood of making good decisions. There are five broad ways in which big data can create value:

- Unlocking significant value by making data transparent/usable at much higher frequency.
- Organizations can collect more accurate/detailed performance data on everything, exposing variability and boosting performance.
- Companies can conduct controlled experiments for better management decisions and/or basic low-freq. forecasting to high-freq. now-casting to adjust their business levers in time.
- Ever-narrower customers' segmentation and more precisely tailored products and services.
- Sophisticated analytics to substantially improve decision-making.
- Improve the development of the next generation of products and services.

Big data is a key basis of competition and growth for individual firms. In most industries, established competitors and new entrants alike leverage data-driven strategies to innovate, compete, and capture value from deep and up-to-real-time. The use of big data can:

- underpin new waves of productivity growth and consumer surplus.
- allow retailers to increase its operating margin by more than 60 percent.
- offer considerable benefits to consumers, companies and organizations through personal-location data, which allows consumers to capture $600 billion in economic surplus.

- **Learning**

Supervised, unsupervised, and semi-supervised are different ways to learn from the data.

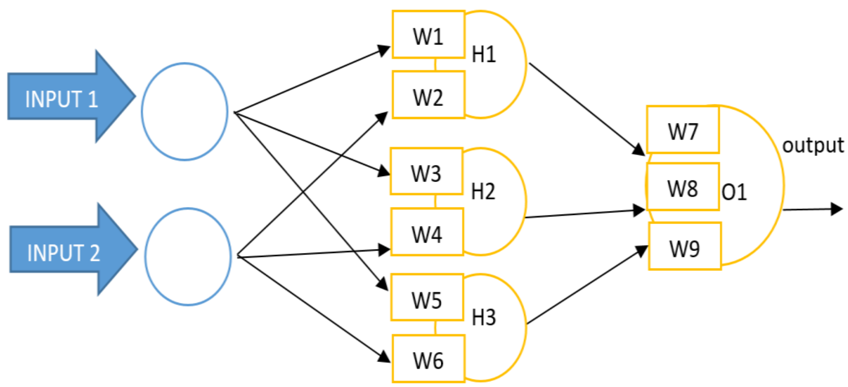
$\boxed{y = f(x)}$ Supervised learning: takes input variables (x) and an output variable (y) and uses an algorithm to learn the mapping function from the input to the output. The goal is to approximate the mapping function so well that you can predict the output variables (y) when you have new input data (x). The process of an algorithm learning from the training dataset can be thought of as a teacher supervising the learning process. We know the correct answers, the algorithm iteratively makes predictions on the training data and is corrected by the teacher. Learning stops when the algorithm achieves an acceptable level of performance. Supervised learning problems can be grouped into regression and classification.

Unsupervised learning: takes input data (x) and no corresponding output variables. The goal is to model the underlying structure or distribution in the data to learn more about the data, and there are no correct answers nor teacher. Algorithms are left to their own devises to discover and present the interesting structure in the data. These problems can be grouped into clustering and association problems. Clustering discovers the inherent groupings in the data, such as grouping customers by purchasing behavior. Association discovers rules that describe large portions of your data, such as people that buy x also tend to buy y. Some popular examples of unsupervised learning algorithms are k-means and Apriori.

Semi-supervised learning: takes a large amount of input data (x) and only some of the data is labeled (y). These problems sit in between both supervised and unsupervised learning. Many real world machine learning problems fall into this because unlabeled data is cheap and easy to collect and store. Unsupervised learning techniques can be used to discover and learn the structure in the input variables, and supervised learning techniques to make best guess predictions for the unlabeled data. Feed the data back into the supervised learning algorithm as training data and use the model to make predictions on new unseen data.

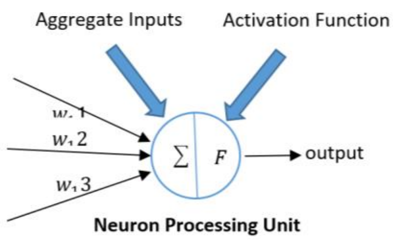
- **Deep Learning**

Neural Network (NN) Architecture:

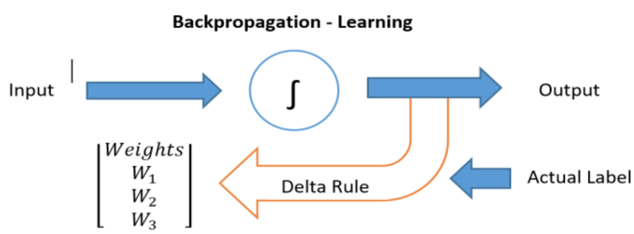


The standard NN architecture: It has an input layer where features are fed in, an output layer with one neuron per class, and a hidden layer of neurons.

In general, all neurons are identical in structure and contain a sum unit and a function unit. In a neural network architecture each input, hidden and output layers has processing unit(s).



Each of the processing unit has inputs along with their weights (for assigning different level of importance to inputs). The inputs along with weights are aggregated and appropriate transformation (activation function) is applied to the aggregated input. Based on the sample data, the weights have to be estimated. There are various algorithms to training the network. One of the commonly used neural network algorithm is *Backpropagation.*

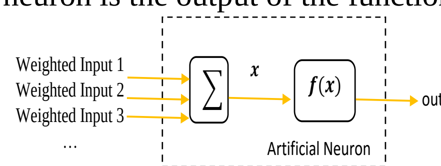


Using initial weights, the target state/labels are estimated using the neural network. The estimate labels are compared against target or actual labels.

The gap is called Error and the error is back propagated to improve the weights and reduce the error.

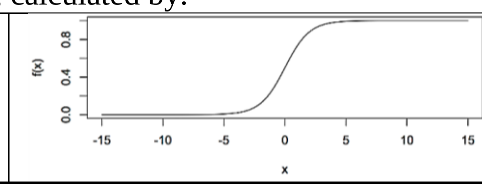
Neural Network Calculation:

Neuron inputs are summed to give a single value, x. This is the input to the function, and the output of the neuron is the output of the function, $f(x)$.



There are a number of functions which can be used in neurons, including the *Radial Basis Function* or a *simple linear function*. The most common function is the *sigmoid or logistic function*, calculated by:

$$f(x) = \frac{1}{1+e^{-x}}$$



This function, as shown in the image above, will always give a value between 0 and 1 so the output of a neuron can only ever be in the range of 0 and 1. Therefore, the result of each output neuron can be calculated in the following manner:

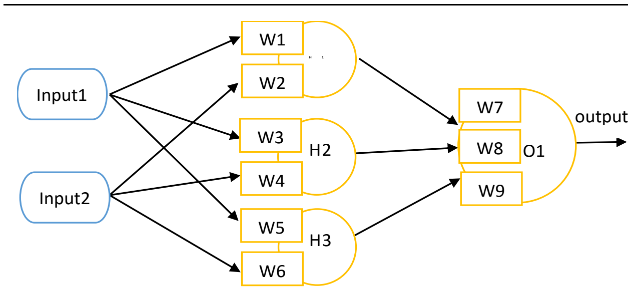
$0_1 = f(w7.h1 + w8.h2 + w9.h3)$

Where $h1, h2,$ and $h3$ can each in turn be calculated:

$h1 = f(w1.i1 + w2.i2)$

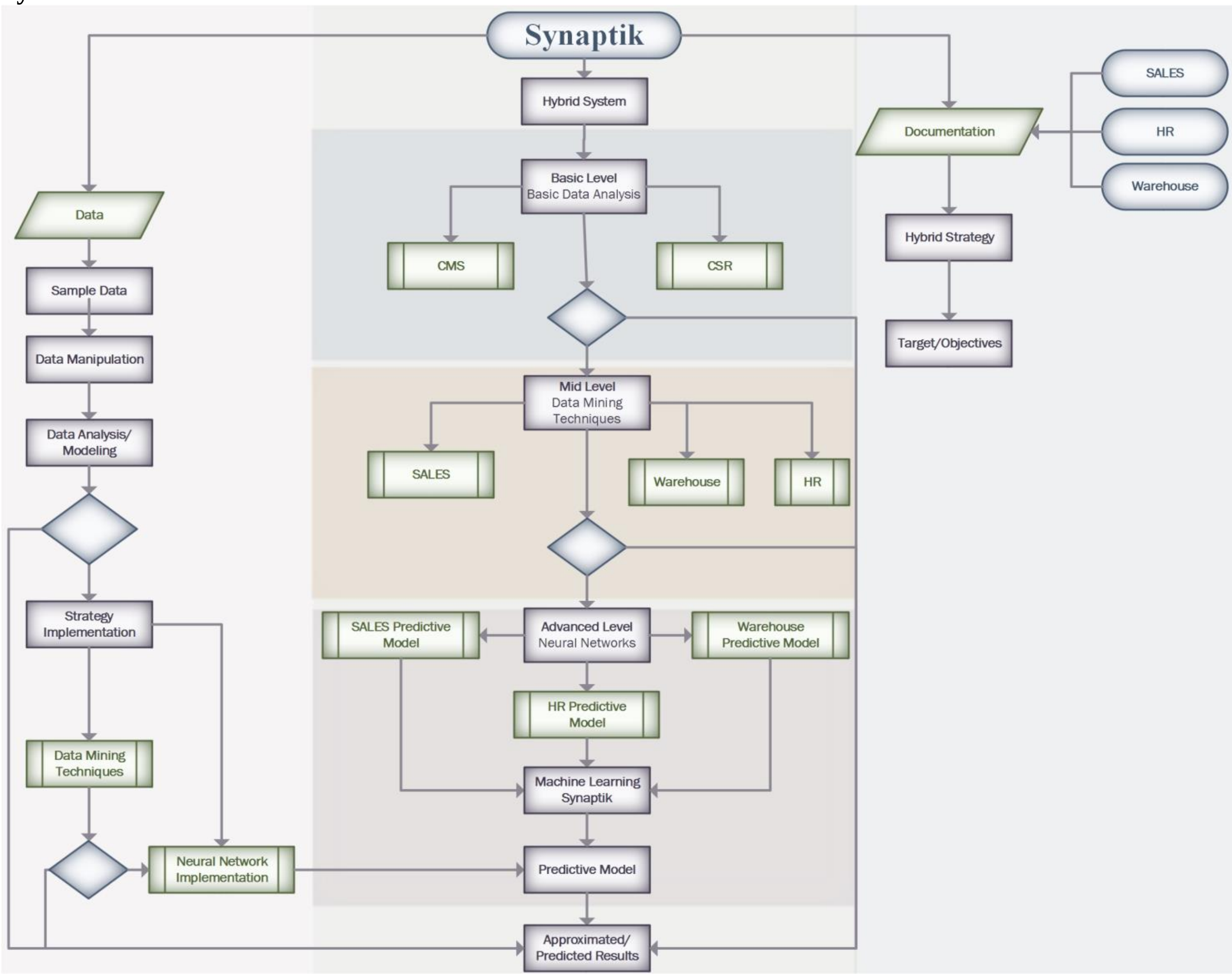
$h2 = f(w3.i1 + w4.i2)$

$h3 = f(w5.i1 + w6.i2)$



This example is for a network that has a single output neuron, making it a binary classification. For the proposed system with x number of output neurons, each neuron is allocated to a class. The final output of the network would be x numbers between 0 and 1 since each output neuron contains a *sigmoid function*, the final highest number is the winner, and the final diagnosis is the class associated with that neuron.

2- System Outline



a. Basic-Level (Basic Data Analysis)
The basic level provides customers a series of tasks that allow them to analyze their data and monitor their performance.

b. Mid-Level (Learning)
The mid-level provides customers a series of data mining tasks that allow them to analyze their data, monitor, and make predictions based on a model that learns from the past data.

c. Advanced-Level (Deep Learning)
The advanced-level provide customers with a series of data mining tasks that allow them to analyze their data, monitor their data and performance, make predictions and obtain recommendations from the model that does not only learn from the past data but also correct itself based on prior/other experiences.

3- System Layers

a. Data

The data is the major part of this project. It is based on the data that the system will be able to measure performance and make decisions based on prior experience. The data must go to a process prior to being analyzed, this process can also be known as "Knowledge Discover in Database" or KDD.

- KDD is the process for identifying valid, new, potentially useful and ultimately understandable patterns in data. It consists of nine steps that begin with the development and understanding of the application domain to the action on the knowledge discovered. Data mining is of the steps and the KDD process is basically the search for patterns of interest in a particular representational one form. On the other hand, OLAP tools focus on providing multidimensional data analysis, which is superior to SQL in computing summaries and control cuts across multiple dimensions. OLAP tools are geared towards simplification and support for interactive data analysis, but the goal of KDD tools is to automate the process as possible. It is a solution used in the field of Business Intelligence, which consists of consultations with multidimensional structures that contain summarized data from large databases or transactional systems.

- Data mining is a step in the KDD process of applying data analysis and discovery algorithms that, under acceptable computational efficiency limitations, produce a particular enumeration of patterns (or models) on the data.

b. Application

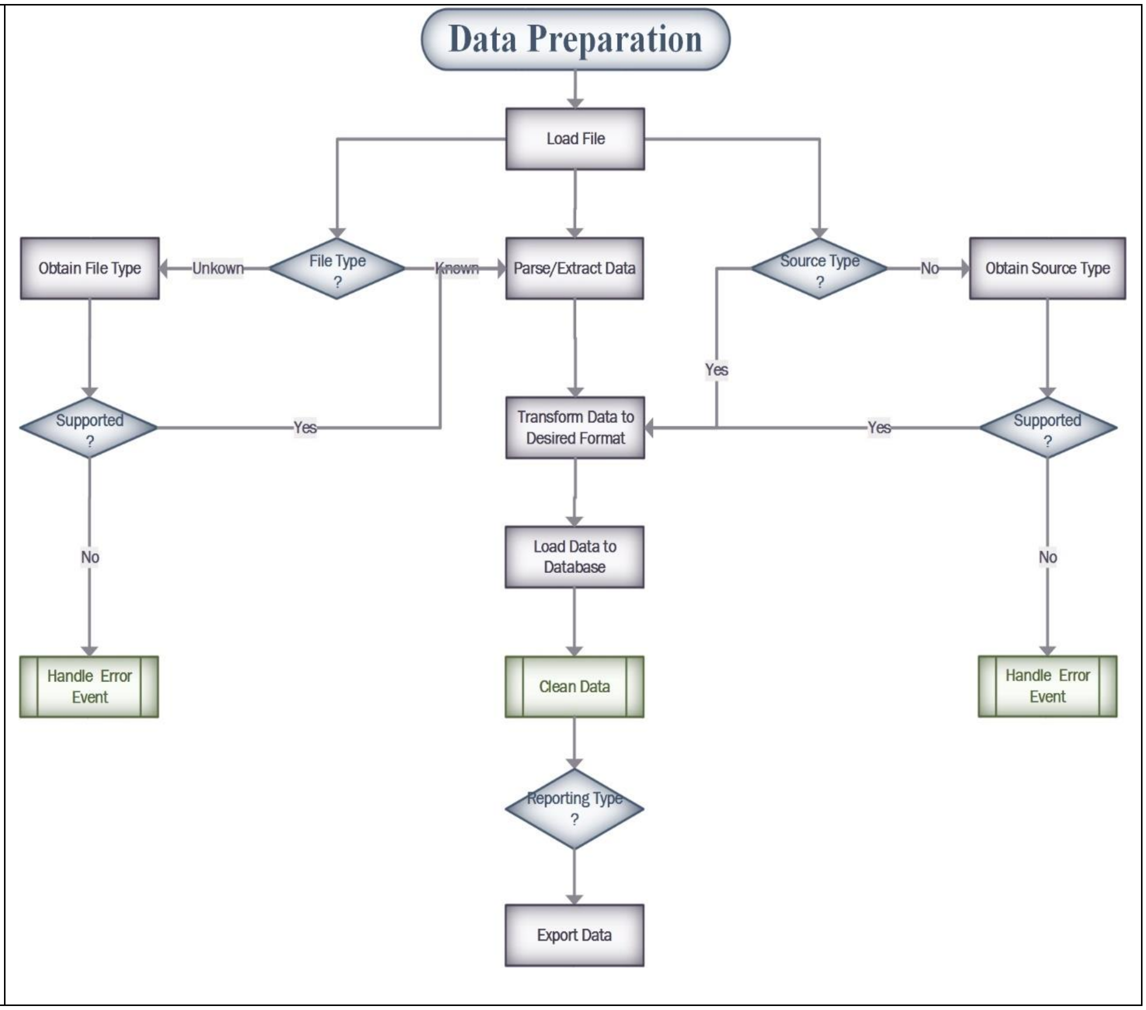
The application setup for this system is currently very simple, and could easily be customized per customer. However, the data application access must be customized by customers because these will provide customers access to data from any application of the field they choose. Not every customer will want to access data from every field but their own. Therefore, data application access has a direct association to the field chosen by the customer. This means that if a customer chooses a field of "warehouse" because of his business, the customer' system will have access to data from applications such as "Warehouse" and so on.

c. Implementation

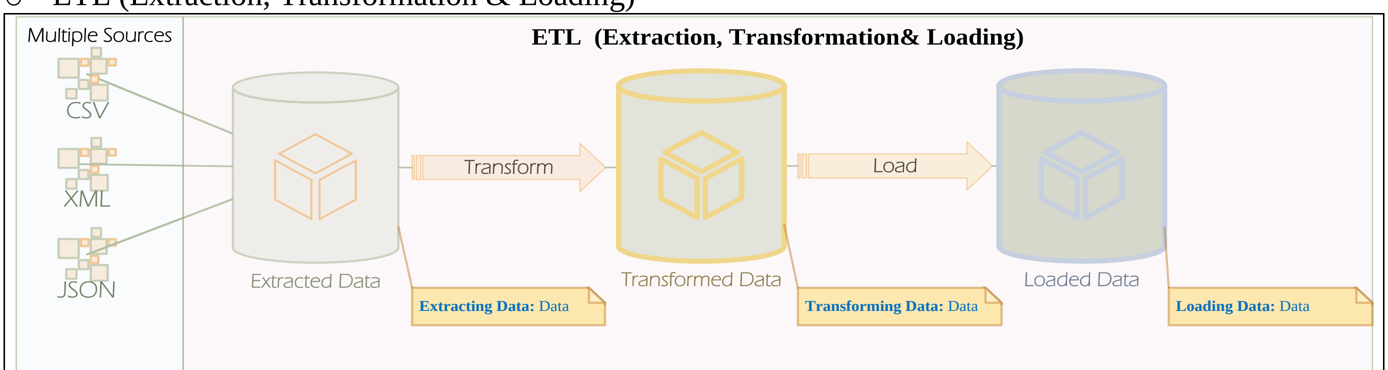
The implementation of the system is the brain, or Synaptic. Once the data is analyzed and models are created for this, every field will have a model that best represents it. Therefore, the implementation of this model will provide customers with predictions and recommendation about their general performance and improvement. The general performance and performance prediction of the customer's business will be solely on the customer's data history; whereas recommendations will solely be based on the customers and all field applications data. This will not only improve the accuracy of general performance prediction for the customers, but it will also provide customers with unknown insights about their business from their own data along with other similar business's data. No private information about any other business will be disclosed, however, the system can easily use the information just for performance and recommendations to improve general performance based on the behavior of the data.

4- Data Layer

- Data Preparation



- ETL (Extraction, Transformation & Loading)



ETL systems integrate data from multiple applications, typically supported by different vendors. The disparate systems containing the original data are frequently operated by different users. Extract, Transform and Load (ETL) refers to a process in database usage and especially in data warehousing that performs:

- Data extraction — extracts data from homogeneous or heterogeneous data sources
- Data transformation — transforms the data for storing it in the proper format for querying and analysis
- Data loading — loads it into the final target (operational data store/data warehouse)

Since the data extraction takes time, it is common to execute the three phases in parallel. While the data is being extracted, another transformation process executes. It processes the already received data and prepares it for loading. As soon as there is some data ready to be loaded into the target, the data loading kicks off without waiting for the completion of the previous phases.

ETL process pulls data out of source systems to place it into Synaptik database through the following tasks:

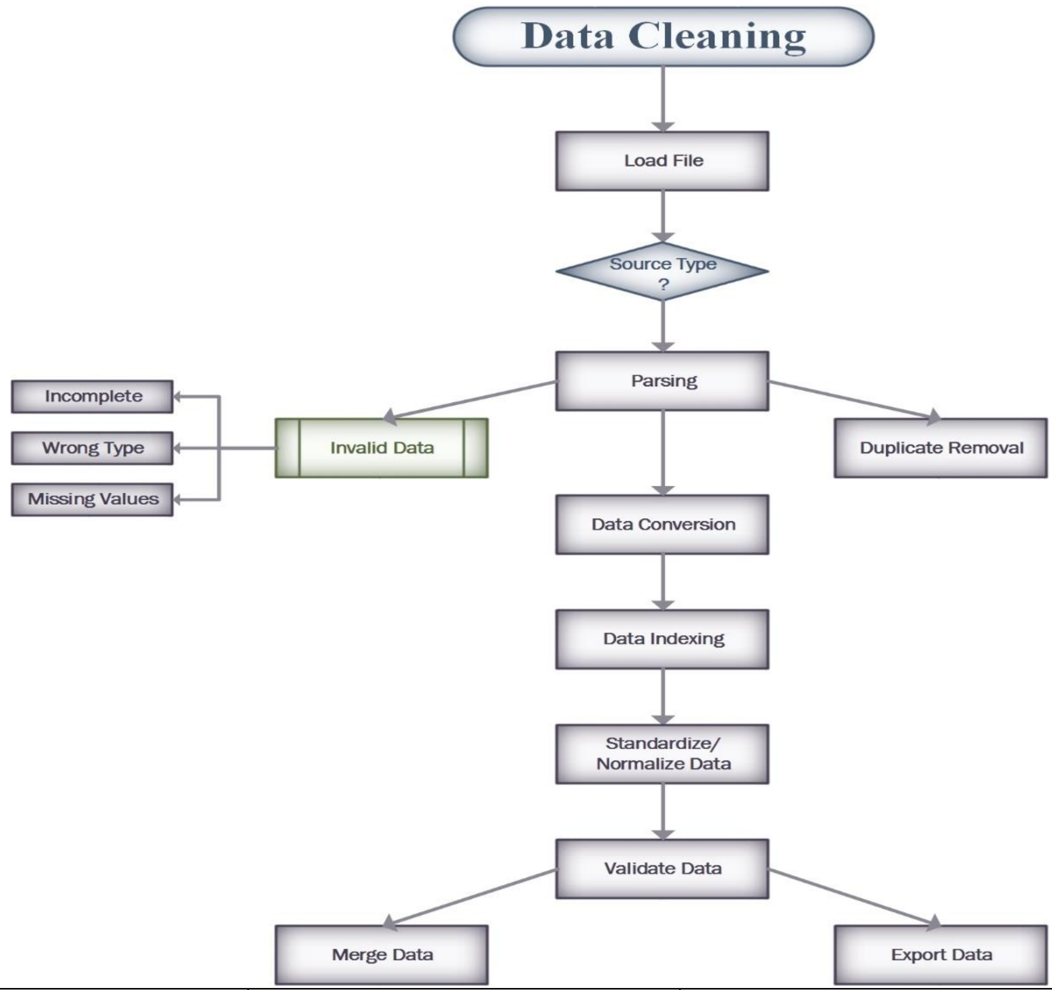
- Extraction
  This is the most important aspect since extracting the data correctly sets the stage for the success of subsequent processes. This system combines data from different source systems, and each system may also use a different data format. Common data-source formats include relational databases, XML and flat files, but may also include non-relational database structures such as Information Management System (IMS) or other data structures such as Virtual Storage Access Method (VSAM) or Indexed Sequential Access Method (ISAM), or even formats fetched from outside sources by means such as web spidering or screen-scraping. In general, it aims to convert the data into a single format(json) appropriate for transformation processing. An intrinsic part of the extraction involves data validation to confirm whether the data pulled from the sources has the correct data type in a given domain. The data is rejected entirely, or in part, if it fails the validation rules. The rejected data is ideally reported back to the source system for further analysis. Sometimes, a data-validation rule is needed to accept data before next phase.

- Transform Data
  In this stage a series of rules are applied to the extracted data to “prepare” it for loading. Some data does not require any transformation, also known as "direct move" data. Cleaning the data is an important function of transformation that aims to pass only "proper" data. The relevant systems' interfacing and communicating is a challenge when different systems interact because of having different characters.

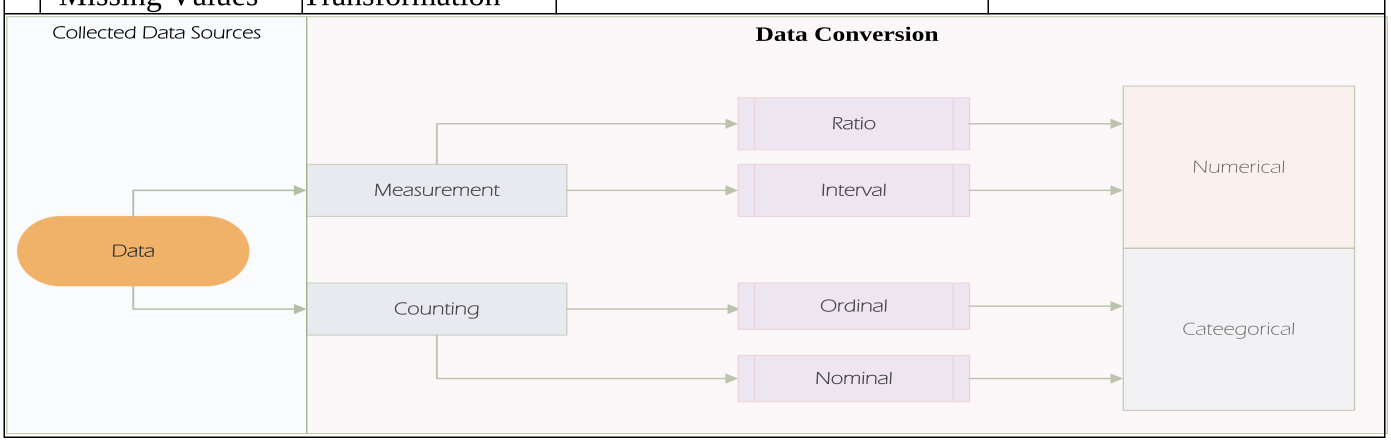
  In some cases, the following transformation types may be required to meet the business and technical needs of the server:
  - Selecting certain columns to load: (or selecting null columns not to load).
  - Translating coded values: (e.g., system codes male as "1" and female as "2")
  - Encoding free-form values: (e.g., mapping "Male" to "M")
  - Deriving a new calculated value: (e.g., sale_amount = qty * unit_price)
  - Sorting the data based on a list of columns to improve search performance
  - Joining data from multiple sources (e.g., lookup, merge) and deduplicating the data
  - Aggregating (for example, rollup — summarizing multiple rows of data — total sales for each store)
  - Generating surrogate-key values
  - Transposing or pivoting (turning multiple columns into multiple rows or vice versa)
  - Splitting a column into multiple columns (e.g., converting a comma-separated list, etc.)
  - Disaggregating repeating columns
  - Looking up and validating the relevant data from tables or referential files
  - Applying any form of data validation; failed validation may result in a full rejection of the data, etc.

- Load Data
  This phase loads the data into the data warehouse. The data warehouses (or other parts of it) may add new data in a historical form at regular intervals. To understand this, consider a data warehouse that is required to maintain sales records of the last year. The entry of data for any one-year window is made in a historical manner. The timing and scope to append are strategic design choices dependent on the time available and the business needs. This system can maintain a history and audit trail of all changes to the data loaded in the data warehouse. As the load phase interacts with a database, the constraints defined in the database schema — as well as in triggers activated upon data load — apply (for example, uniqueness, referential integrity, mandatory fields), which also contribute to the overall data quality performance of the ETL process.

- Data Cleaning

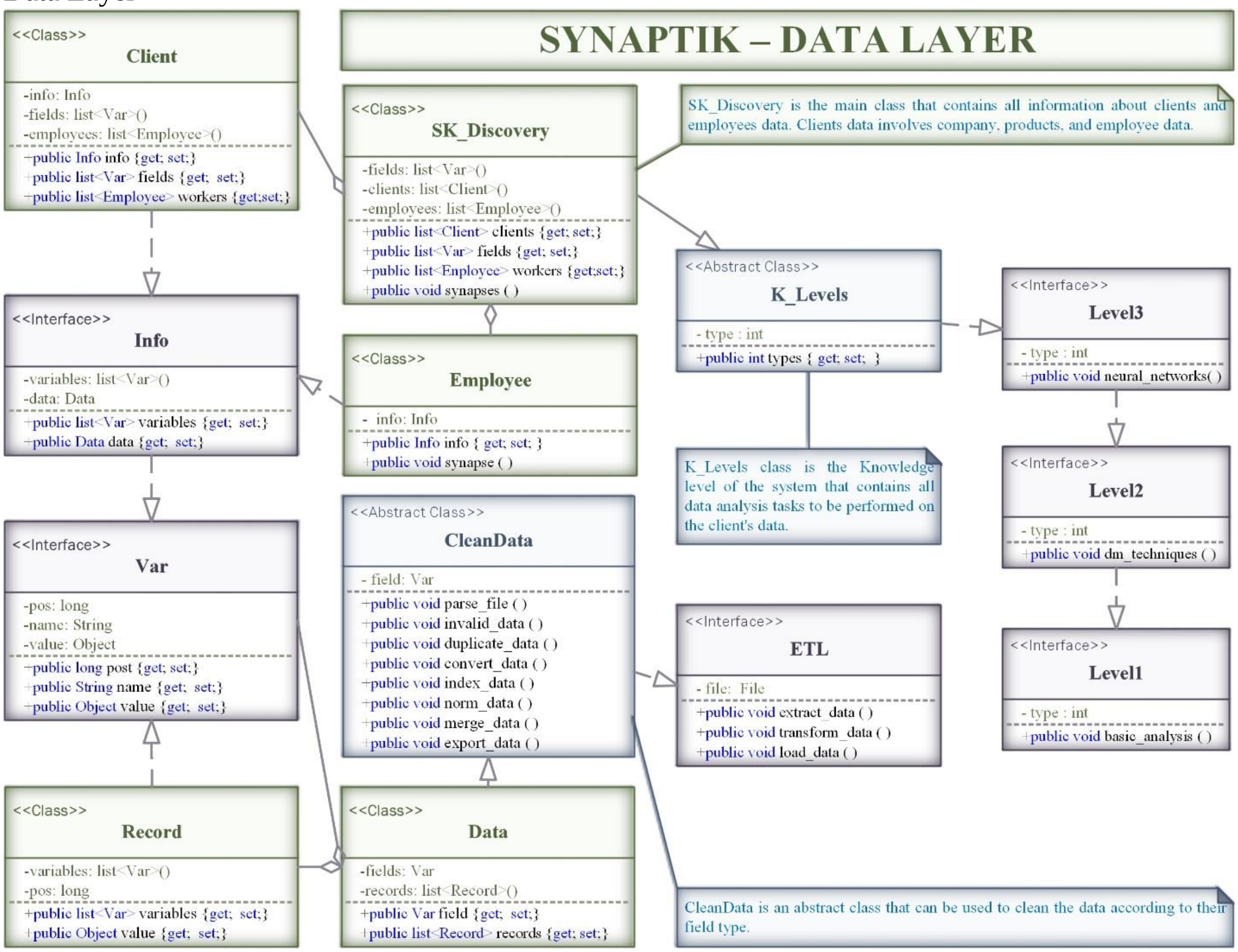


	Parsing - Invalid Data - Missing Values	- Wrong Type - Incomplete Data Transformation	- Convert Nominal to Numeric - Indexing	Duplicate Elimination Statistical/Database Methods



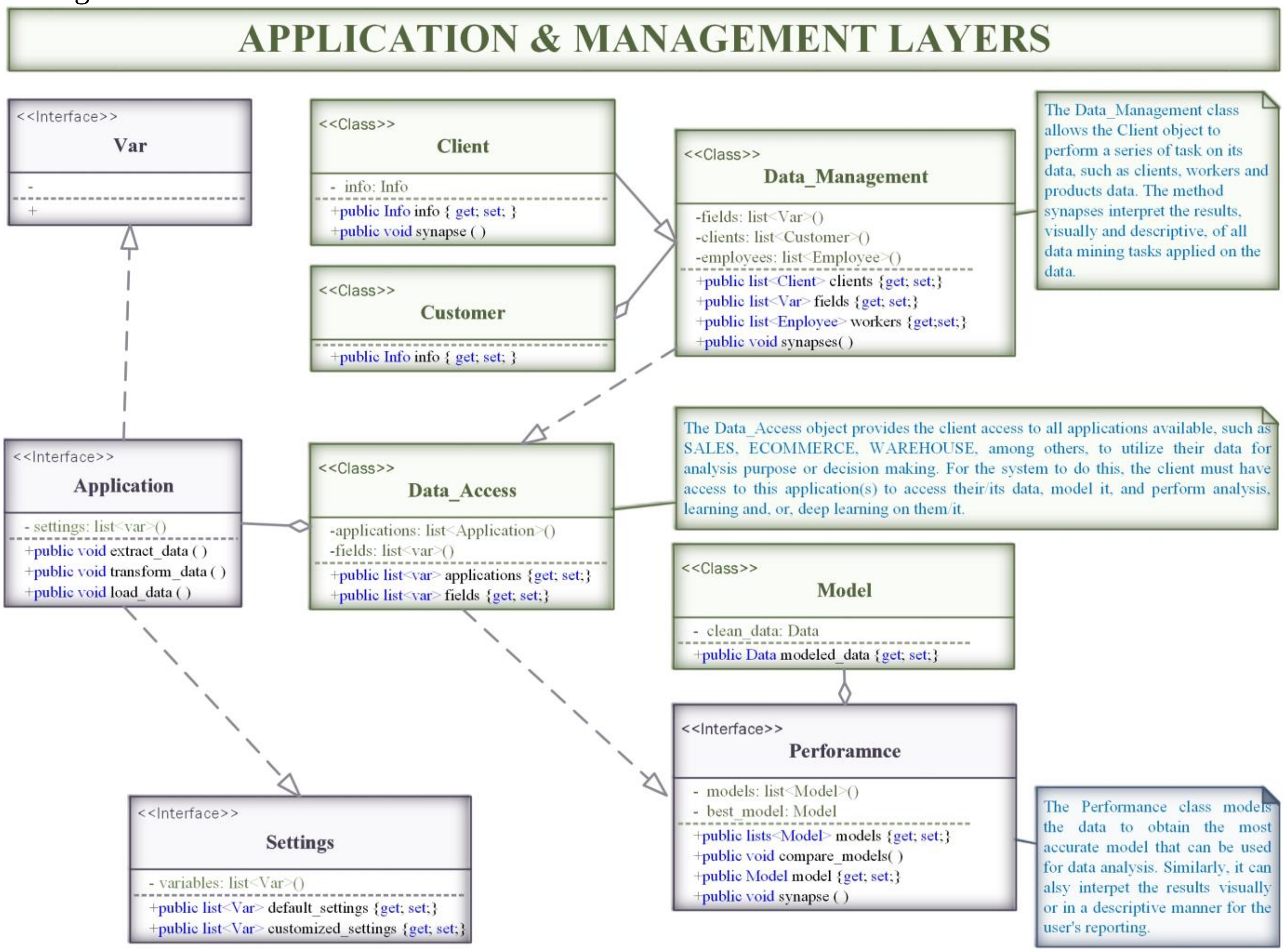
Basic operations include removing noise if appropriate, collect the necessary information to model or account for noise, deciding on strategies for handling missing data fields and account for time sequence information and known changes.

- Database Structure
    - Data Layer



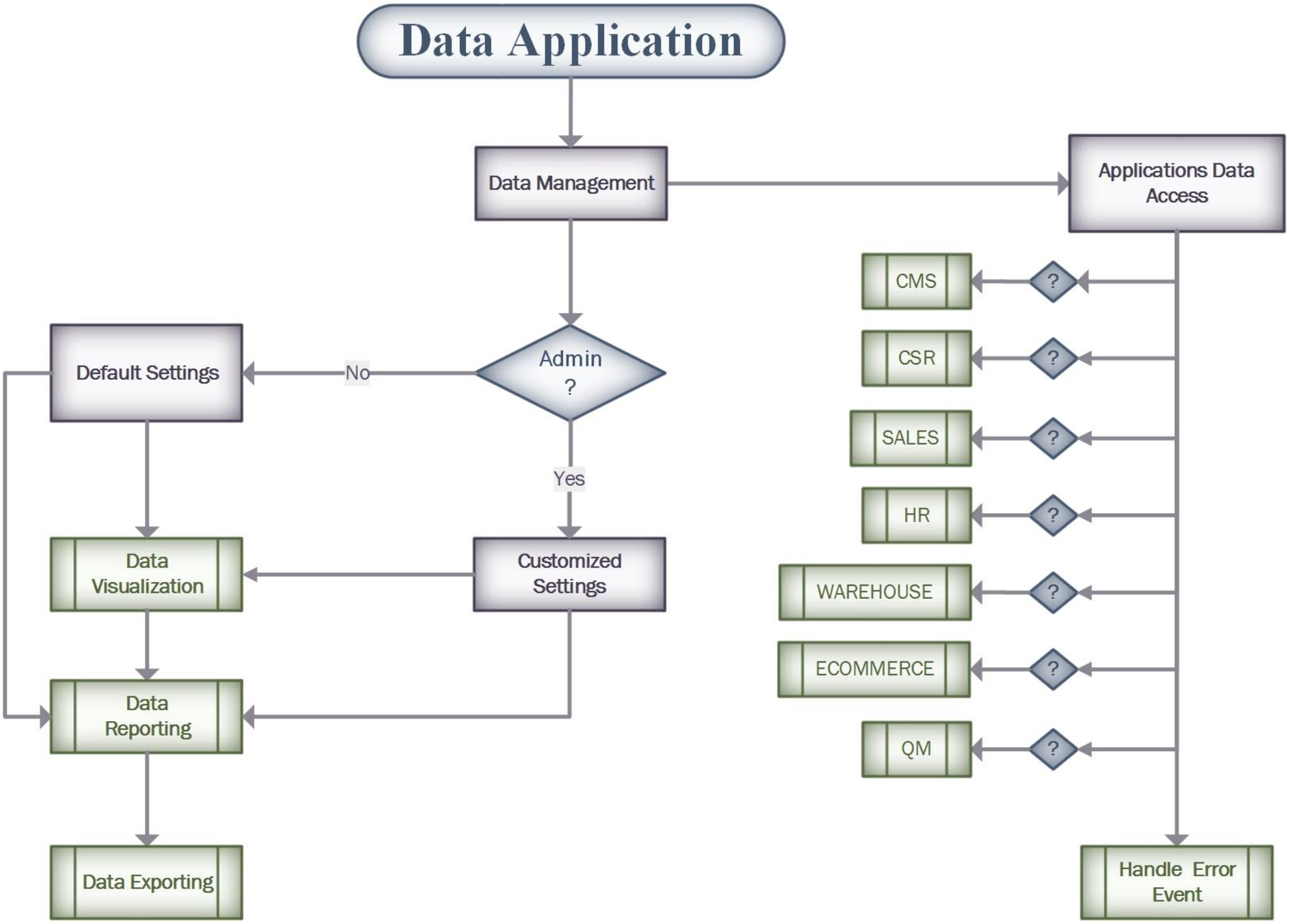
Object Type	Name	Description	Relationships
Interface	Level1	Provide users with Basic Data Analysis functions	None
Interface	Level2	Provide users with Data Mining Tasks for Learning and Data Analysis	Implements Level1
Interface	Level3	Provide users with Advanced Data Mining Tasks for Deep Learning and Data Analysis	Implements Level2
Abstract Class	K_Levels	Provide users with functions to discover knowledge from their data based on their level.	Implements Level3
Class	SK_Discovery	Provide an overall Knowledge Discovery of Synaptik based on what the system has learned from the data behavior of its fields & applications.	Extends K_Levels
Class	Employee	Object to represent an employee from a client or Synaptik	It is an Object of SK_Discovery &
Class	Client	Object to represent a customer of Synaptik	Implements Info
Interface	Info	Provide a list of fields that identify each user	Implements Var
Interface	Var	Enclose the value and name of an abstract object	None
Interface	ETL	Provide all functions to implement the ETL process	None
Abstract Class	CleanData	Provide all functions to clean the data, clean data	Implements ETL
Class	Data	Every record is contained in data object(dataset)	Extends CleanData
Class	Record	Every attribute is contained in a Record	Implements Var

- Manages



Object Type	Name	Description	Relationships
Interface	Var	Enclose the value and name of an abstract object	None
Interface	Settings	Contains selected, or customized, variable for data analysis	None
Interface	Application	ETL process for customer's application data	Implements Var & Settings
Interface	Performance	Performs data analysis and creates a data model	None
Class	Model	Contains the results of all analysis performed on data	Object of Performance
Class	Data_Access	Provide access to all available applications data for analysis purpose such as performance.	Implements Performance
Class	Data_Management	Manage data analysis tasks for clients, customers, and business data.	Extends Data_Access
Class	Customer	Object to represent a client of Customer (Implements Info)	Object of Data _Management

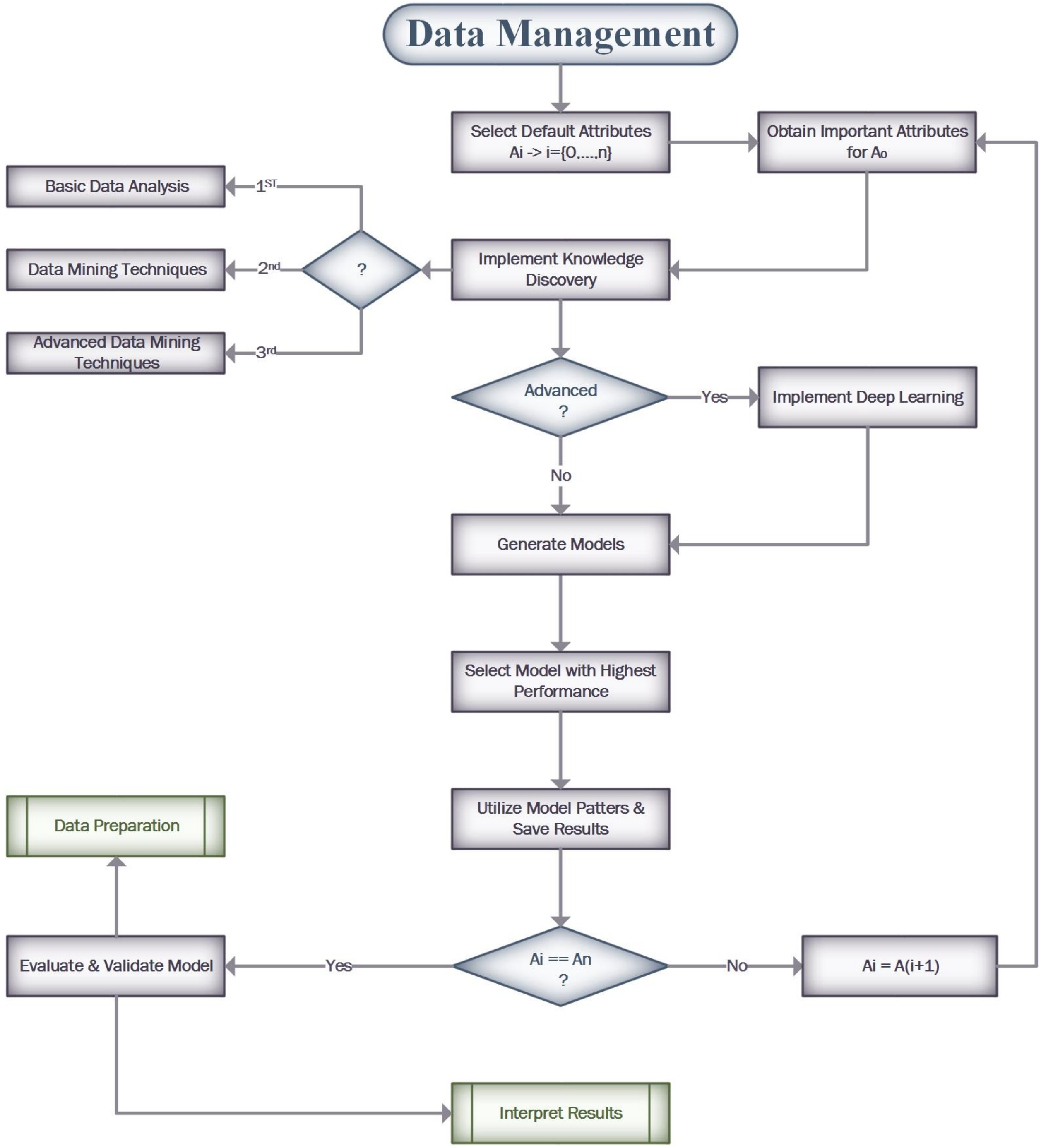
5- Application Layer



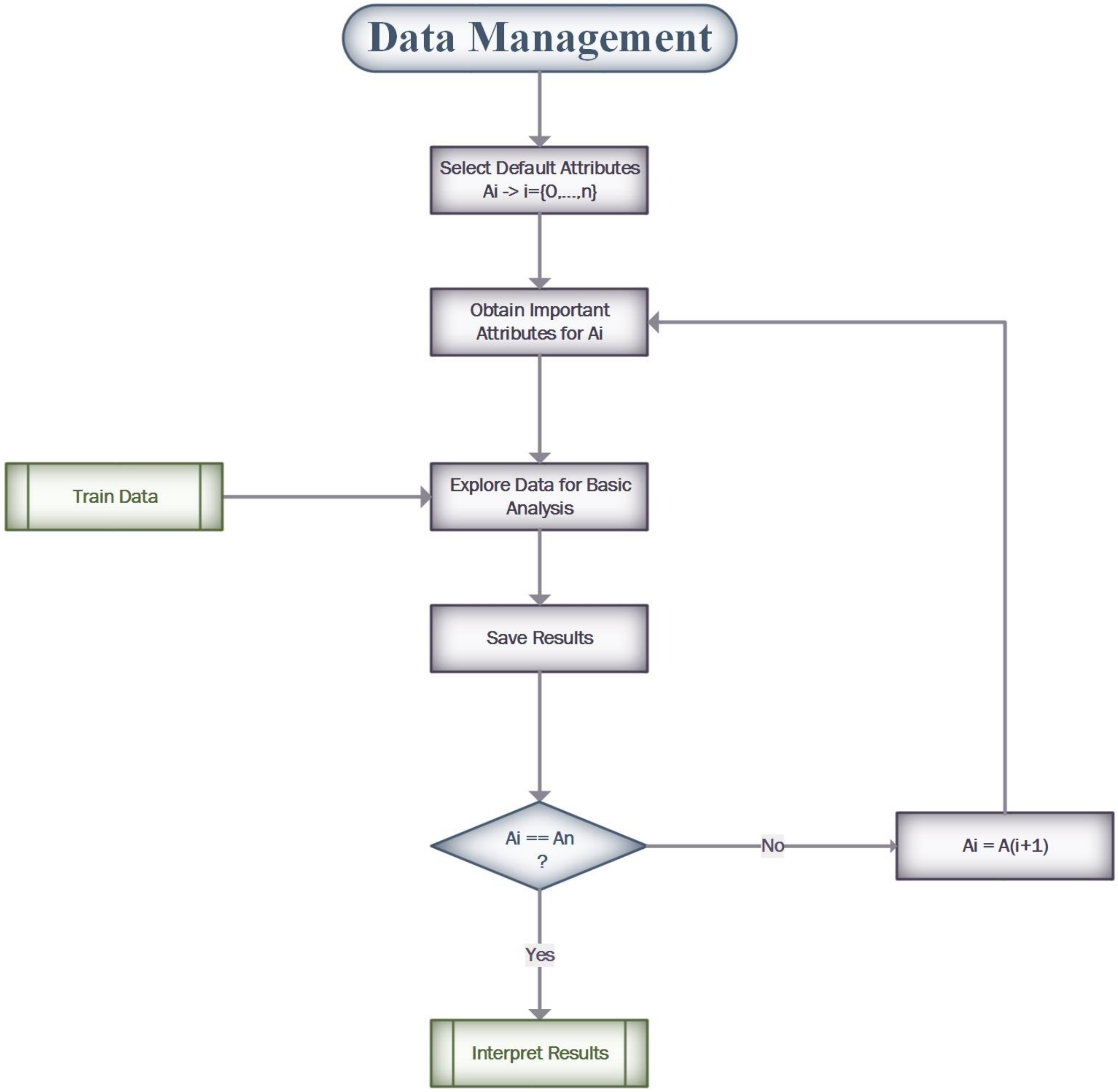
Application Setup

1. Cloud Synaptik Repository Setup
2. Ruby Environment Setup
3. ETL Data Structure
   - . csv
   - . xml
   - . json
   - Others
4. Database setup
   - Hadoop setup
   - ETL & Hadoop Integration

6- Management Layer

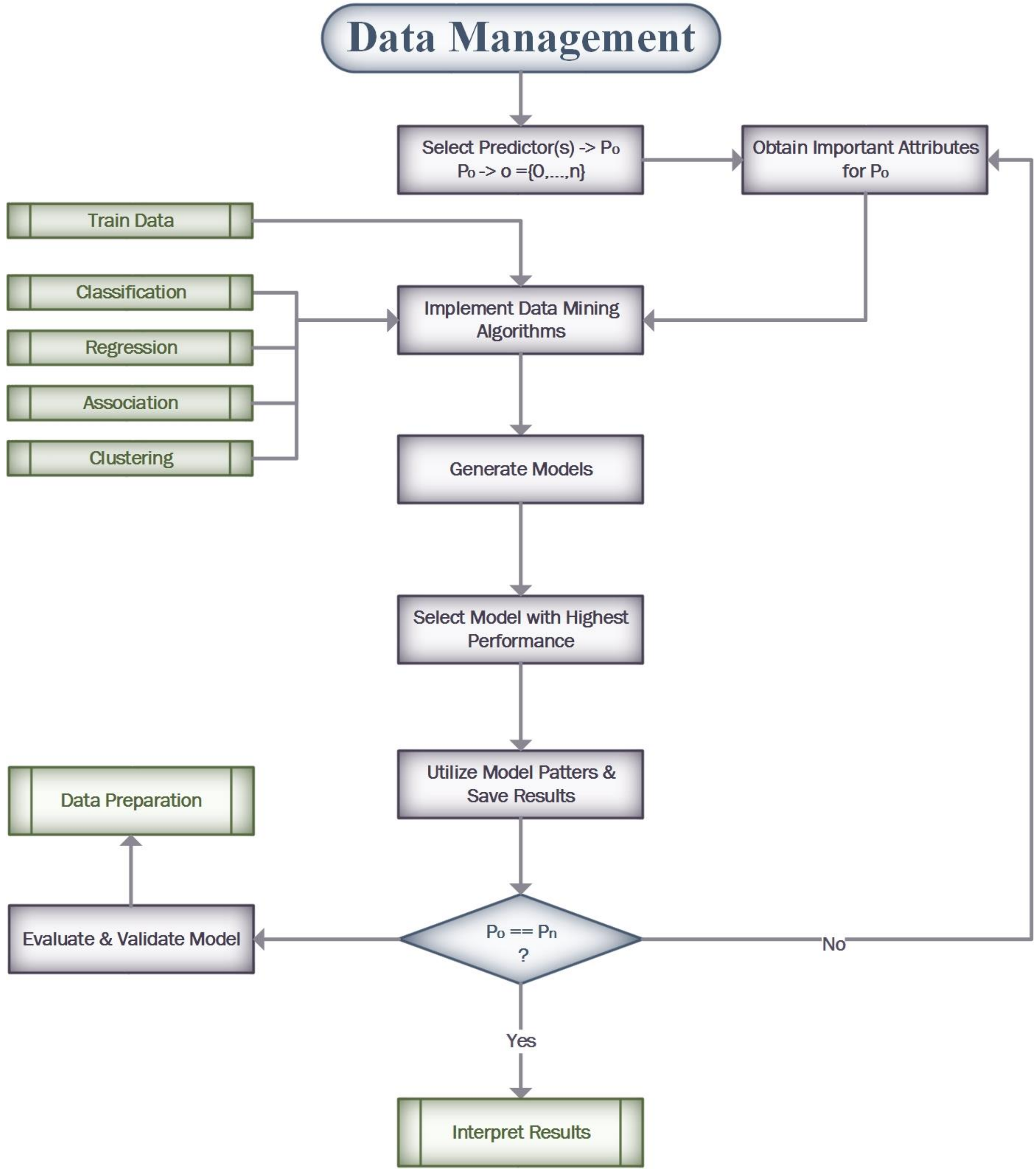


The data management class provides users with tasks to implement knowledge discovery on their database, which is composed of customers, employees and products data. These tasks are performed on the data according to the layer that the user belongs to, either basic data analysis, data mining tasks or advanced data mining tasks.



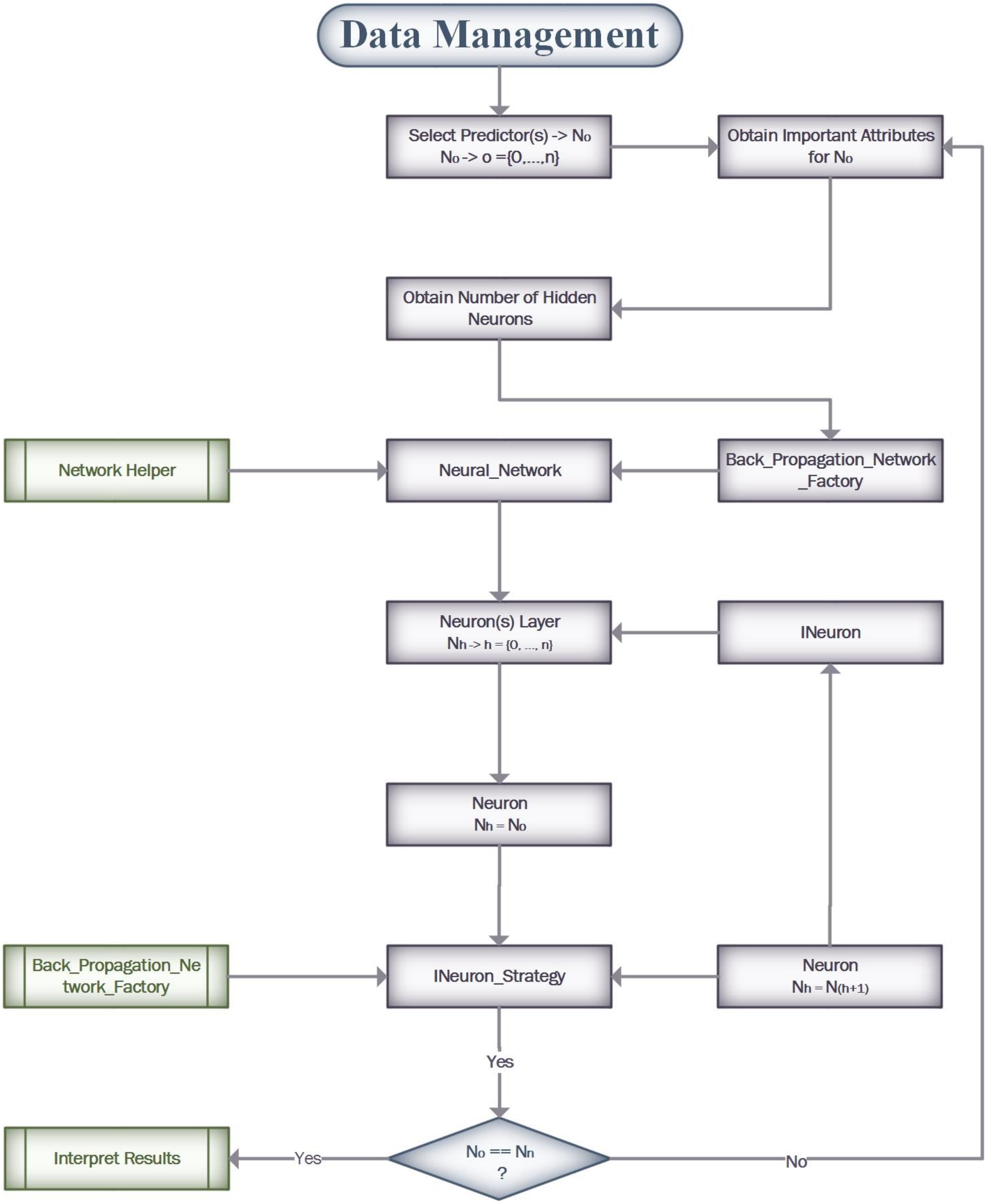
Level 1

At this level, basic data analysis is performed to explore the data and monitor the performance of selected attributes within the data given data. The goal is to find patterns that disclose unknown insights about the data and the selected attributes performance.



Level 2

At this level, data mining techniques are performed on the data to learn from past trends and data behavior. The machine learning is created after applying the first layer to explore the data and find the state, or performance, of the selected attributes. The goal is to find patterns that disclose unknown insights about the data and selected attributes performance while learning from the history, trends and behavior of the past data to predict the state of selected attributes in a given future or scenario.



Level 3

At this level, advanced data mining techniques are performed on the data to deeply learn about the data through a neural network. This level is implemented after level 2, which provides the system with knowledge learned from past trends and behavior of the data after exploring it and finding the state of the selected attributes. The goal is to find patterns that disclose unknown insights about the data and selected attributes performance while learning from the history, trends and behavior of the past data. In this manner, the system is able to predict the state of selected attributes in a given time or scenario through a Neural Network, and find problems and solutions within the data that can help improve the performance in real time. In fact, it can automatically detect errors within the data and make modifications appropriately based on the behavior of the data, among other factors, and provide users with a series of recommendations